Hierarchical Multimodal Memory for Training-Free Video Moment Retrieval

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Abstract

Video moment retrieval grounds a natural-language query in a temporal span of a video. Long and repeated-query settings benefit from boundary-sensitive localization, reusable memory, and auditable evidence. We introduce Hierarchical Multimodal Memory Module (HMMM), a training-free pipeline that builds multimodal memory from visual captions and ASR transcripts, then retrieves moments with saliency-based watershed proposals and multi-signal re-ranking. On the QVHighlights validation subset with available frames, HMMA outperforms Moment-DETR on 6 of 7 metrics and exposes multimodal evidence for each prediction.

CCS Concepts

• **Information systems → Multimedia and multimodal retrieval**; **Video search**.

Keywords

video moment retrieval, multimodal retrieval, hierarchical memory, temporal localization

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1 Introduction

Task and motivation. Given a video and a text query, video moment retrieval returns the queried event's start and end time [6, 7]. For example, given "A man starts playing guitar," the system must identify both the matching scene and its temporal boundary. This becomes difficult in long videos, where many visually similar events may appear and spoken content can be necessary to disambiguate the target. Recent video large language models (Video LLMs) provide strong video understanding, yet long-form retrieval still faces two practical limits: processing all frames and tokens is costly, and summary-based abstractions are often built for video QA rather than precise temporal boundaries. This has motivated training-free pruning [2, 3, 8], token compression [9, 15], and hierarchical abstraction methods [4, 17], but existing systems typically rely on unimodal entities [4], captions [10], or tree summaries [17] instead of the multimodal evidence and query-conditioned boundaries needed for QVHighlights [11] and TVR-Ranking [12]. Together, these constraints distinguish retrieval from general video QA: the model must preserve query-specific boundary evidence and expose the visual or spoken support for the returned interval.

Core Idea. We propose HMMA, a query-aware pipeline that links fine-grained candidate generation with reusable event memory. It moves beyond caption-derived spans by proposing windows from query saliency and re-ranking them with multimodal hierarchical evidence. This preserves temporal precision while making each retrieved span auditable.

Contributions. We make three contributions: (1) a training-free hierarchical multimodal memory for evidence-grounded retrieval; (2) a saliency-based proposer for multiple boundary candidates; and (3) reusable per-video memory for repeated queries without rebuilding video evidence.

2 Hierarchical Multimodal Memory Pipeline

2.1 Overview

HMMA builds reusable three-level memory once per video and performs query-time retrieval over the cache. Construction segments the video, attaches visual captions and ASR transcripts, and groups segments into events; retrieval proposes query-salient candidate moments and re-ranks them with cached evidence. Encoders remain frozen: we use CLIP ViT-L/14 for image–text matching [13] and MiniLM for caption/query matching [16].

The segmenter run TW-FINCH [14] on per-frame CLIP features, choose the partition closest to eight leaves, and enforce 6–40 s segments. The captioner outputs GPT-4o-mini [1] captions from four keyframes and Whisper-large-v3 ASR fragments routed by temporal overlap for each segment. The memory builder stores segment spans, captions, ASR, and MiniLM embeddings at Level 1; groups Level-1 embeddings into up to four Level-2 event nodes with average-linkage clustering, storing centroids and temporal hulls; and stores a Level-3 video centroid for future cross-video gating. At query time, a watershed-style proposer extracts windows around saliency peaks under five thresholds. Candidates are scored by CLIP frame-query similarity, saliency coverage, caption match, ASR match, and event-centroid match, then fused by Reciprocal Rank Fusion [5] with k=60: CLIP/coverage weights are 1.0, caption/ASR weights are 0.3, the Level 2 weight is selected by ablation, and temporal NMS at 0.5 IoU keeps the top 10 windows.

3 Experiments

3.1 Setup

QVHighlights val [11] provides 1,137 scored queries; our frame-based pipeline uses the 1,111/1,519 videos with available frames. We compare the trained **Moment-DETR** baseline with three training-free variants: **Ours-Vision** uses watershed proposals with visual saliency and CLIP matching; **Ours-Vision-Cap** adds visual-caption matching; and **Ours-HMMA** adds ASR and Level 2 event-centroid.

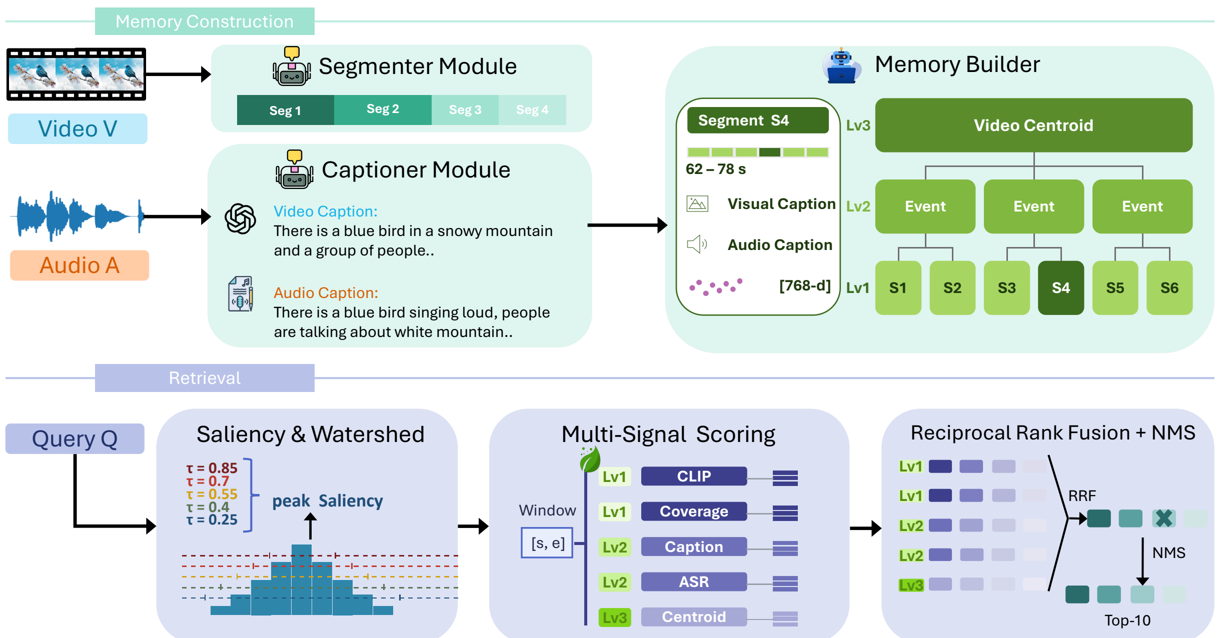

Figure 1: Overview of HMMA. A reusable Hierarchical Multimodal Memory is built once per video from segment captions, ASR transcripts, and event clusters; query-relevant moments are retrieved by watershed proposals and multi-signal re-ranking.

Table 1: QVHighlights val (1,137 queries). MR/HL denote moment retrieval/highlight detection; scores are percentages, best bold.

Metric	Moment DETR	Ours Vision	Ours Vision-Cap	Ours HMMA
MR-R1@0.5	53.94	53.12	55.15	**55.85**
MR-R1@0.7	**34.84**	33.25	34.65	34.65
MR-mAP	30.73	**36.17**	34.98	35.15
MR-mAP@0.5	54.82	57.97	59.27	**59.92**
MR-mAP@0.75	29.40	**36.08**	33.41	33.50
HL-mAP	35.64	67.05	67.05	**67.05**
HL-Hit@1	55.60	74.32	74.32	**74.32**

3.2 Main Results

Table 1 shows that **Ours-HMMA** beats Moment-DETR on 6/7 metrics. The largest gains are HL-mAP (+31.4 pp) and HL-Hit@1 (+18.7 pp). MR-mAP@0.5 rises by +5.1 pp; Ours-Vision raises MR-mAP@0.75 by +6.7 pp.

Ablations. Table 2 shows that $w^\star$=0.2 gives the best R1@0.5 and mAP@0.5, while $w^\star$=0.3 gives the best R1@0.7 and overall mAP; both improve over removing the Level 2 signal ($w^\star$=0.0).

Cost. Memory construction costs about 10 s per video on A6000 GPUs, mostly from Whisper-large-v3 ASR, plus about \$5$\times10^{-4}$ for GPT-4o-mini captions. Cached queries run in about 25 s on one GPU without repeated video processing or API calls.

Table 2: Ablation of the Level-2 cluster weight $w^\star$ on HMMA.

$w^\star$	R1@0.5	R1@0.7	mAP	mAP@0.5	mAP@0.75
0.00	55.15	34.39	35.17	59.59	33.62
0.05	55.32	34.39	35.08	59.60	33.51
0.10	55.50	34.48	35.12	59.73	33.52
0.15	55.58	34.74	35.20	59.84	**33.67**
0.20	**55.85**	34.65	35.15	**59.92**	33.50
0.30	55.58	**34.92**	**35.23**	59.81	33.66
0.50	54.35	34.04	34.85	59.27	33.22

4 Discussion and Conclusion

HMMA shows that a training-free multi-module system can match or exceed a specialized trained model when query-conditioned watershed candidates are paired with hierarchical multimodal memory. Beyond performance, the system offers an evidence-grounded retrieval interface: returned spans are supported by saliency, visual captions, ASR transcripts, and event-level memory rather than a single opaque score. This is important for long videos, where an incorrect temporal span can misrepresent an extended recording and where repeated queries should reuse the same video evidence without rebuilding it. Separating memory construction from retrieval is practical: captions and ASR are cached once, while later queries can still adjust candidate windows and evidence ranking. In our results, watershed proposals improve boundary-sensitive localization and Level-2 memory adds event context for moderate-IoU retrieval. Future work will broaden long-video validation [12] and extend Level-2/3 gating to cross-video retrieval.

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